1010 quiz1

1. An experiment was conducted to assess the effect of baking temperature on the density of bread. A large batch of white bread dough and a large batch of whole wheat bread dough were prepared. Each of the batches was formed into 18 loaves, resulting in a total of 36 loaves. Each of the white bread loaves was randomly assigned to one of three baking temperatures (low, medium, and high), so that 6 loaves were baked at the low temperature, 6 were baked at the medium temperature, and 6 were baked at the high temperature. A similar process of random assignment was carried out for the whole wheat bread loaves.

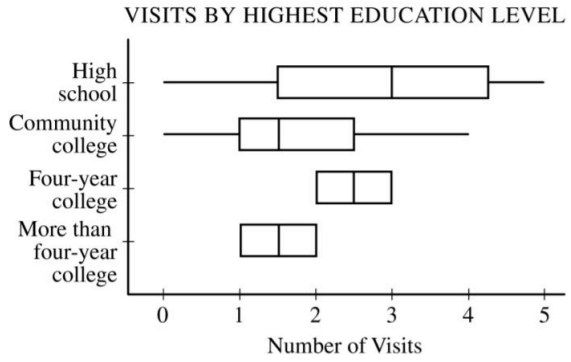
Which of the following are the treatments in the experiment?

(A) The 36 loaves of bread
(B) The type of bread (white, whole wheat)
(C) The density of the bread
(D) The baking temperatures (low, medium, high)

2. For the purpose of determining the value of its end-of-year inventory, a clothing store creates a list at the end of the year of every item currently in stock along with each item's wholesale price. Which of the following is the best description of the end-of-the-year activity?

(A) An experiment, because the items are treatments and wholesale prices are responses.
(B) A sample survey, because the store wants to estimate the value of all items for the entire year.
(C) A sample survey, because the items currently in the store at the end of the year are a random sample of all items in the store for the entire year.
(D) A census, because the wholesale prices of all items are listed.

3. A sociologist collected data from a sample of people on their highest level of education and the number of times they visited any fast food restaurant during the previous week. The data are summarized in the boxplots.

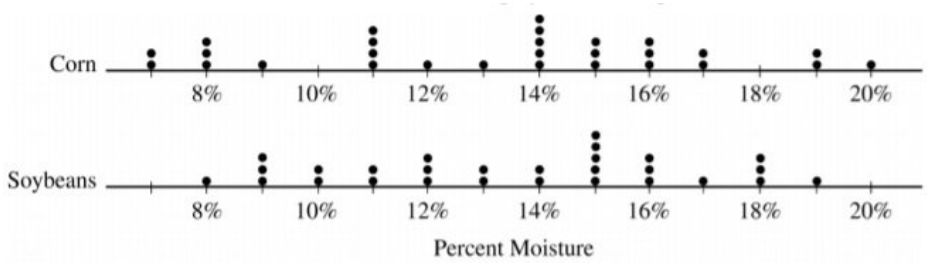


Based on the boxplots, which of the following statements must be true?

(A) The proportion of people surveyed from the first quartile to the third quartile at the four-year college level is less than the respective proportion at the community college level.
(B) The interquartile range (IQR) for the number of visits at the more than four-year college level is less than the IQR for the number of visits at the community college level.
(C) The maximum number of visits at the community college level is greater than the maximum number of visits at the high school level.
(D) The median number of visits at the four-year college level is greater than the median number of visits at the high school level.

1010 quiz1

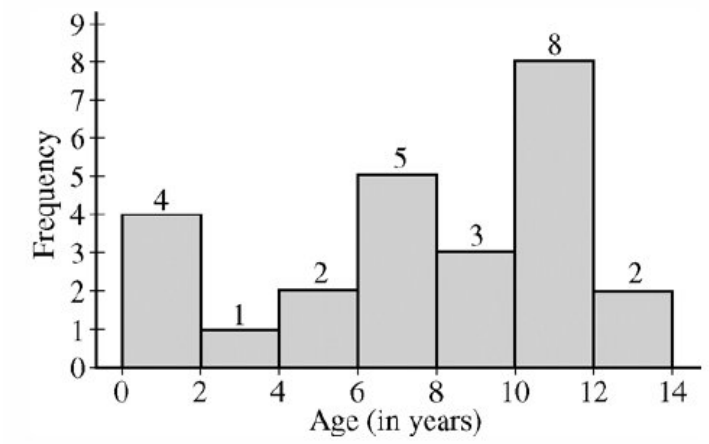
4. Grain moisture is a characteristic of grain that affects the price paid for the grain. A random sample of 28 loads of corn was evaluated for moisture as a percent of the total weight. A different random sample of 28 loads of soybeans was also evaluated for moisture. The data are displayed in the dotplots below.



Based on the dotplots, which of the following is greater for the percent moisture of corn than for the percent moisture of soybeans?

(A) The first quartile
(B) The median
(C) The third quartile
(D) The range

5. A random sample of 25 households from the Mountainview School District was surveyed. In this survey, data were collected on the age of the youngest child living in each household. The histogram below displays the data collected in the survey.



In which of the following intervals is the median of these data located?

(A) 4 years old to less than 6 years old
(B) 6 years old to less than 8 years old
(C) 8 years old to less than 10 years old
(D) 10 years old to less than 12 years old

6. For a recent season in college football, the total number of rushing yards for that season is recorded for each running back. The mean number of rushing yards for the running backs that season is 790 yards. One running back had 1,637 rushing yards for the season, which is 2.42 standard deviations above the mean number of rushing yards.

What is the standard deviation of the number of rushing yards for the running backs that season?

(A) 250 yards

(B) 300 yards
(C) 350 yards
(D) 400 yards

7. A compact disc (CD) manufacturer wanted to determine which of two different cover designs for a newly released CD will generate more sales. The manufacturer chose 70 stores to sell the CD. Thirty-five of these stores were randomly assigned to sell CDs with one of the cover designs and the other 35 were assigned to sell the CDs with the other cover design. The manufacturer recorded the number of CDs sold at each of the stores and found a significant difference between the mean number of CDs sold for the two cover designs. Which of the following gives the conclusion that should be made based on the results and provides the best explanation for the conclusion?

(A) It is **not** reasonable to conclude that the difference in sales was caused by the different cover designs because this was not an experiment.
(B) It is **not** reasonable to conclude that the difference in sales was caused by the different cover designs because there was no control group for comparison.
(C) It is **not** reasonable to conclude that the difference in sales was caused by the different cover designs because the 70 stores were not randomly chosen.
(D) It is reasonable to conclude that the difference in sales was caused by the different cover designs because the cover designs were randomly assigned to stores.

8. For a sample of 42 rabbits, the mean weight is 5 pounds and the standard deviation of weights is 3 pounds. Which of the following is most likely true about the weights for the rabbits in this sample?

(A) The distribution of weights is approximately normal because the sample size is 42, and therefore the central limit theorem applies.
(B) The distribution of weights is approximately normal because the standard deviation is less than the mean.
(C) The distribution of weights is skewed to the right because the least possible weight is within 2 standard deviations of the mean.
(D) The distribution of weights is skewed to the left because the least possible weight is within 2 standard deviations of the mean.

9. The number of hurricanes reaching the East Coast of the United States was recorded for each of the last ten decades by the National Hurricane Center. Summary measures are shown below.

Min = 12	Max = 24
Lower quartile = 15	Upper quartile = 18
Median = 16	n = 10

Which of the following statements is true?

(A) The largest observation is 24 and it is an outlier. No other observations in the data set could be outliers.
(B) Both 12 and 24 are outliers. It is possible that there are also other outliers.
(C) 12 is an outlier and it is possible that there are other outliers at the low end of the data set. There are no outliers at the high end of the data set.
(D) 24 is an outlier and it is possible that there are other outliers at the high end of the data set. There are no outliers at the low end of the data set.

10. Administrators at a state university computed the mean GPA (grade point average) for juniors and seniors majoring in either physics or chemistry. The results are displayed in the table below. When juniors and seniors are grouped together, could physics majors have a higher mean GPA than chemistry majors?

1010 quiz1

	Physics	Chemistry
Juniors	2.8	3.0
Seniors	3.2	3.6
Overall	?	?

(A) No. The physics majors' mean GPA for juniors and seniors must be 3.0, while the chemistry majors' mean GPA for juniors and seniors must be 3.3.

(B) No. There is not enough information to determine the mean GPA for each major, but it must be higher for chemistry majors than for physics majors.

(C) Yes. It could happen. Whether it does happen depends on the number of juniors and seniors in each major.

(D) Yes. It could happen. Whether it does happen depends on the variability of the GPAs within each of the four groups of students.

11. A data set of test scores is being transformed by applying the following rule to each of the raw scores.

$$\text{Transformed score} = 3.5(\text{raw score}) + 6.2$$

Which of the following is NOT true?

(A) The mean transformed score equals 3.5(the mean raw score) + 6.2.

(B) The range of the transformed scores equals 3.5(the range of the raw scores) + 6.2.

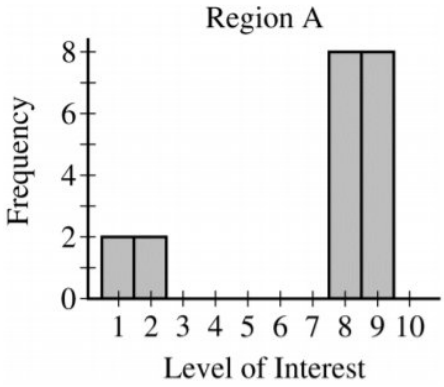
(C) The standard deviation of the transformed scores equals 3.5(the standard deviation of the raw scores).

(D) The IQR of the transformed scores equals 3.5(the IQR of the raw scores).

12. A marketing firm obtained random samples of 20 people in five regions of the country to investigate the level of interest in a new product. People in the sample were asked to rate their level of interest on a scale from 1 to 10, with 1 being the least amount of interest and 10 being the greatest. The histograms show the results for each region.

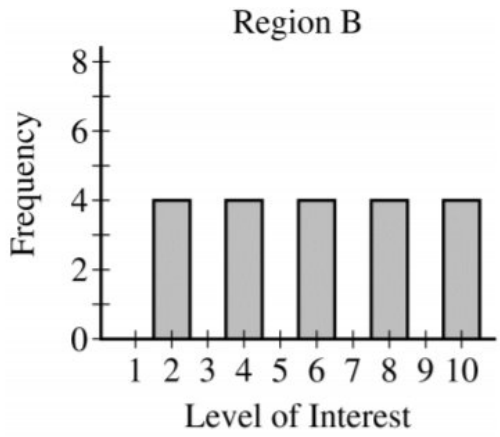
The graph for which region displays data for level of interest with the least standard deviation?

(A)

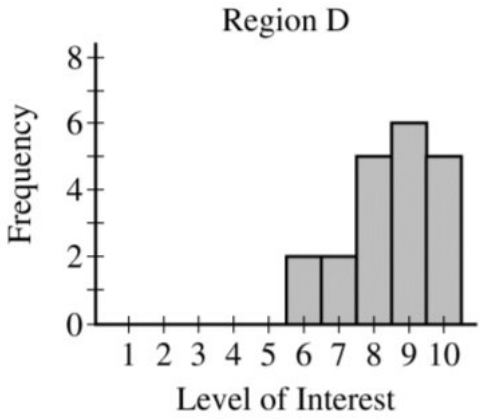


1010 quiz1

(B)

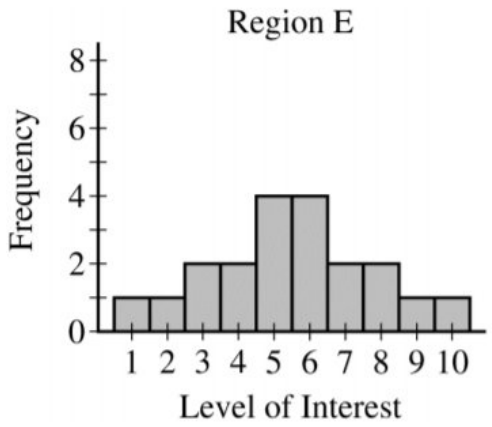


(C)



1010 quiz1

(D)



13. A new restaurant is interested in determining the best time-temperature combination for roasting a five-pound cut of lamb. The times to be tested are 45 minutes, 60 minutes, and 90 minutes at temperatures of 350 degrees Fahrenheit and 425 degrees Fahrenheit for each time, with the exception of the 90 minute - 425 degree combination. That combination is being eliminated because it will overcook the lamb, which leaves five combinations remaining. From 10 identical cuts of lamb, 2 are randomly selected to roast using each of the time-temperature combinations in the same oven. The quality of the finished product is evaluated for each roast. Which of the following is true?

(A) The explanatory variable is the quality of the finished lamb.
(B) If the experiment is repeated, identical results will be expected.
(C) There should be a control group (i.e., a group in which no treatment is given).
(D) The two cuts that are being roasted for each time-temperature combination are an example of replication.

14. The final scores for 309 participants in a national golf tournament are summarized in the following table.

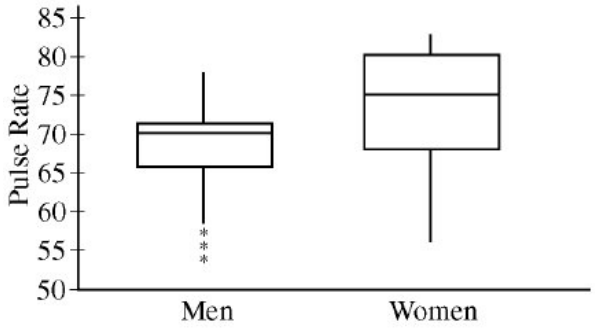
N	Mean	SE Mean	St Dev	Min	Q1	Median	Q3	Max
309	149.35	0.37	6.42	131	145	150	153.5	168

Which of the following is closest to the z-score of the minimum golf score?

(A) -2.16
(B) -2.86
(C) -2.96
(D) -18.40

15. The pulse rate for each person in a sample of 20 men and 20 women was recorded. The boxplots below summarize the pulse rates for the men and the women in the sample.

1010 quiz1



Which of the following statements about the people in the sample must be true?

(A) There are more people between the first and third quartiles for women than there are between the first and third quartiles for men.
(B) At least half of the women had higher pulse rates than three-fourths of the men.
(C) More than half of the men had lower pulse rates than three-fourths of the women.
(D) If a man and a woman were randomly selected from the 40 people, the man would have the lower pulse rate.

16. An athletic trainer believes that a new natural treatment will decrease the mean healing time, in days, compared to the current treatment for athletes with knee injuries. The athletic trainer would like to create a study to test his belief. The mean healing time for athletes receiving the current treatment is 21 days. An investigative question will be formulated before the study is created.

Which of the following would be the appropriate components of the athletic trainer's investigative question?

(A) The parameter is the mean healing time for knee injuries when using the new natural treatment, and the direction of the alternative hypothesis is not equal to 21.
(B) The parameter is the mean healing time for knee injuries when using the new natural treatment, and the direction of the alternative hypothesis is less than 21.
(C) The parameter is the healing time for knee injuries when using the new natural treatment, and the direction of the alternative hypothesis is not equal to 21.
(D) The parameter is the healing time for knee injuries when using the new natural treatment, and the direction of the alternative hypothesis is less than 21.

17. Heights, in inches, for the 200 graduating seniors from Washington High School are summarized in the frequency table below.

Height	Frequency
Height < 60 inches	22
60 inches $\leq$ height < 66 inches	84
66 inches $\leq$ height < 72 inches	62
72 inches $\leq$ height < 78 inches	24
Height ≥ 78 inches	8

Which of the following statements about the median height is true?

(A) It is greater than or equal to 72 inches but less than 78 inches.
(B) It is greater than or equal to 66 inches but less than 72 inches.
(C) It is greater than or equal to 60 inches but less than 66 inches.

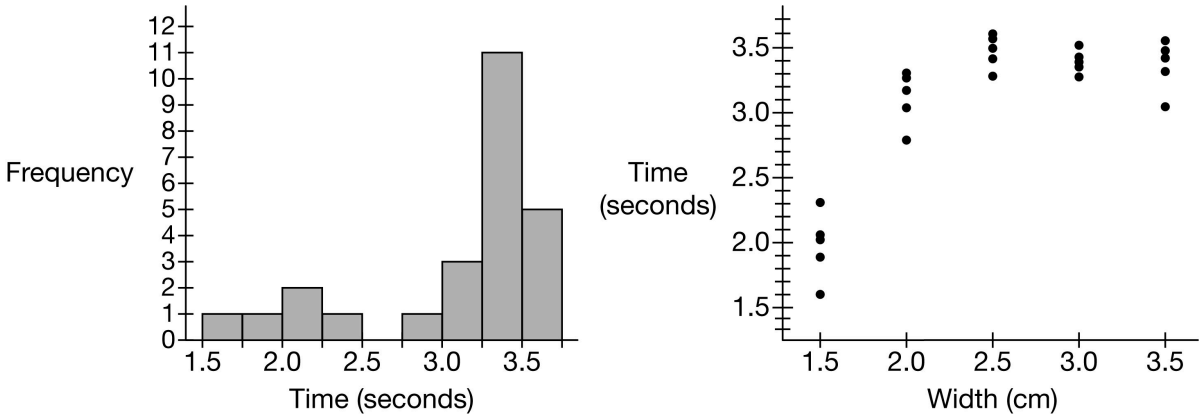
(D) It is less than 60 inches.

18. A doctor wanted to publish a research study about newborn babies in a certain city. She randomly selected newborn babies from the city and recorded their weights, in kilograms, using the same scale. She later discovered that the scale she used was incorrectly adding 0.25 kilogram to the weight of each newborn baby. To correct the error, she adjusted each of the weights in the random sample by subtracting 0.25 kilogram.

Which of the following statistics for the weights, in kilograms, of the newborn babies has the same value before and after the doctor's adjustment?

(A) Median
(B) Mean
(C) Third quartile
(D) Interquartile range

19. Students in a physics class designed an experiment to investigate the effect of blade width on flight time for paper helicopters. A histogram of the flight times, in seconds, and a scatterplot of flight time versus blade width, in centimeters (cm), are shown for 25 helicopters.



Which of the following best describes the clusters and the gap in the histogram?

(A) The left cluster displays flight times for blade widths from 1.5 cm to 2.5 cm, and the right cluster displays flight times for the remaining blade widths.
(B) The left cluster displays flight times for blade widths of 1.5 cm, and the right cluster displays flight times for the remaining blade widths.
(C) The clusters are not related to the blade widths because flight time is measured in seconds and blade widths are measured in centimeters.
(D) The clusters are not related to the blade widths because the relationship between blade width and flight time is not linear.

20. A study was conducted to evaluate the impact of taking a nutritional supplement on a person's reaction time. One hundred volunteers were placed into one of three groups according to their athletic ability: low, moderate, or high. Participants in each group were randomly assigned to take either the nutritional supplement or a placebo for six weeks. At the end of the six weeks, participants were given a coordination task. The reaction time in completing the task was recorded for each participant. The study compared the reaction times between those taking the supplement and those taking the placebo within each athletic ability level.

Which of the following is the best description of the study?

(A) A randomized block design
(B) A completely randomized design
(C) A matched-pairs design

1010 quiz1

(D) A randomized observational study

21. The transportation department of a large city wants to estimate the proportion of residents who would use a system of aerial gondolas to commute to work. The gondolas would be part of the city's effort to relieve traffic congestion. The department asked a random sample of residents whether they would use the gondolas. The residents could respond with yes, no, or maybe.

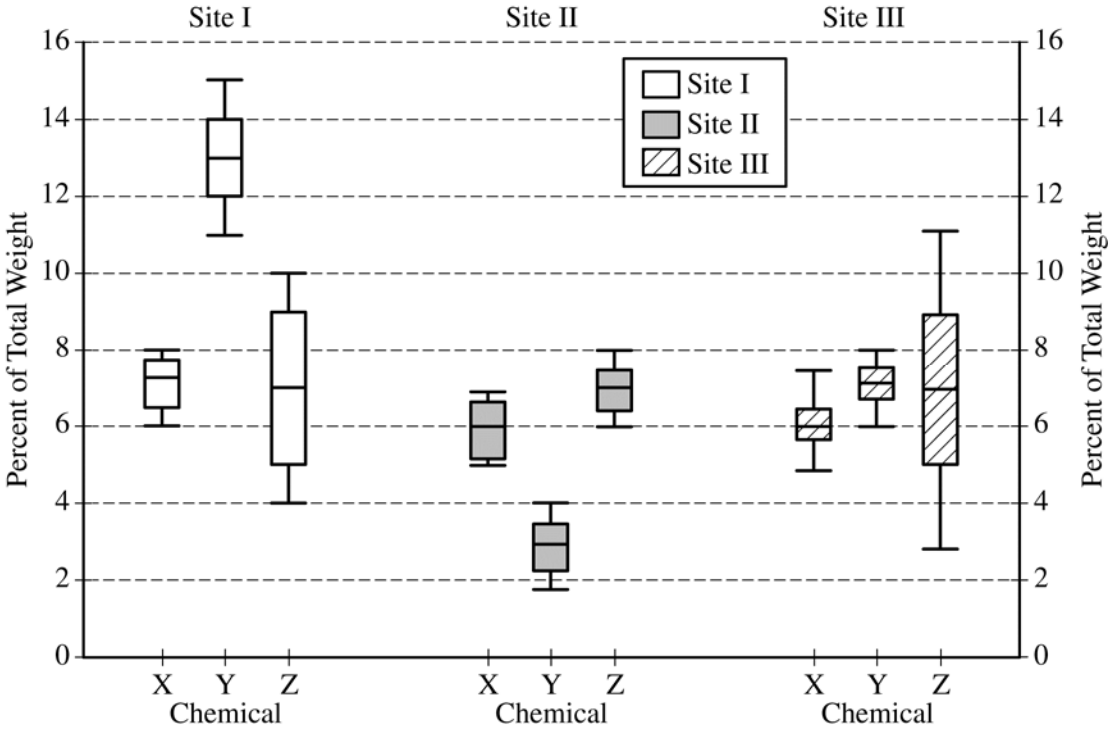
Which of the following is the best description of the method for data collection used by the department?

(A) A census
(B) A sample survey
(C) An experiment with a completely randomized design
(D) An experiment with a randomized block design

22. A researcher wanted to estimate the average amount of money spent on extracurricular activities per school in a certain region. The researcher randomly selected 20 public schools and 20 private schools in the region to use for a sample. Which of the following best describes the type of sample that was taken?

(A) A cluster sample
(B) A convenience sample
(C) A simple random sample
(D) A stratified random sample

4. The chemicals in clay used to make pottery can differ depending on the geographical region where the clay originated. Sometimes, archaeologists use a chemical analysis of clay to help identify where a piece of pottery originated. Such an analysis measures the amount of a chemical in the clay as a percent of the total weight of the piece of pottery. The boxplots below summarize analyses done for three chemicals—X, Y, and Z—on pieces of pottery that originated at one of three sites: I, II, or III.



(a) For chemical Z, describe how the percents found in the pieces of pottery are similar and how they differ among the three sites.

(b) Consider a piece of pottery known to have originated at one of the three sites, but the actual site is not known.

(i) Suppose an analysis of the clay reveals that the sum of the percents of the three chemicals X, Y, and Z is 20.5%. Based on the boxplots, which site—I, II, or III—is the most likely site where the piece of pottery originated? Justify your choice.

(ii) Suppose only one chemical could be analyzed in the piece of pottery. Which chemical—X, Y, or Z—would be the most useful in identifying the site where the piece of pottery originated? Justify your choice.

The median percent of chemical Z is about 7% at all three sites [①]. However, the variability of chemical Z differs across the sites [②]: Site II has the smallest range (about 6% to 8%, range ≈ 2%), Site I has a range of about 6% (4% to 10%), and Site III has the largest range (about 3% to 11%, range ≈ 8%). [③ total weight]

© 2017 The College Board.
Visit the College Board on the Web: www.collegeboard.org.

> **Chemical Y** would be the most useful **[选 Y]**. The boxplots of chemical Y for the three sites **do not overlap**: Site I is about 11%–15%, Site II is about 1.9%–4%, and Site III is about 6%–8%. So any measured percent of Y matches only one site. In contrast, the boxplots of chemicals X and Z overlap a lot across the three sites, so they cannot tell the sites apart **[理由：同时谈位置和变异]**.

- **E**：选 Y，且理由说明 Y 的分布在三地**互不重叠、各不相同**（同时考虑位置和变异）。
- **P**：选 Y，理由基于箱线图但没讲清“不重叠”；或者讲清了 Y 不重叠、但没有明确选 Y。

2014 AP® STATISTICS FREE-RESPONSE QUESTIONS

4. As part of its twenty-fifth reunion celebration, the class of 1988 (students who graduated in 1988) at a state university held a reception on campus. In an informal survey, the director of alumni development asked 50 of the attendees about their incomes. The director computed the mean income of the 50 attendees to be $189,952. In a news release, the director announced, "The members of our class of 1988 enjoyed resounding success. Last year's mean income of its members was $189,952!"

   (a) What would be a statistical advantage of using the median of the reported incomes, rather than the mean, as the estimate of the typical income?

   (b) The director felt the members who attended the reception may be different from the class as a whole. A more detailed survey of the class was planned to find a better estimate of the income as well as other facts about the alumni. The staff developed two methods based on the available funds to carry out the survey.

   Method 1: Send out an e-mail to all 6,826 members of the class asking them to complete an online form. The staff estimates that at least 600 members will respond.

   Method 2: Select a simple random sample of members of the class and contact the selected members directly by phone. Follow up to ensure that all responses are obtained. Because method 2 will require more time than method 1, the staff estimates that only 100 members of the class could be contacted using method 2.

   Which of the two methods would you select for estimating the average yearly income of all 6,826 members of the class of 1988 ? Explain your reasoning by comparing the two methods and the effect of each method on the estimate.

The median is less affected by skewness and outliers than the mean **[①]**. Incomes are likely **skewed to the right**: a few class members with very large incomes (for example, someone earning $5,000,000 a year) would pull the mean up a lot but would barely change the median **[②]**. So the median gives a better estimate of a typical income.

I would choose **Method 2 [选 M2]**. Method 1 relies on **voluntary response**: only 600 of the 6,826 members chose to reply to the e-mail **[S2 ①]**. Members with higher incomes are probably more willing to share their income than members with lower incomes, so the people who respond likely have different incomes from those who don't **[S3 ①]**. As a result, Method 1 would likely **overestimate** the mean yearly income of the class **[S3 ②]**. Method 2 uses a **simple random sample** and follows up until every selected member responds **[S2 ②]**, so even with a smaller sample size it is more representative and gives an unbiased estimate.

Begin your response to **QUESTION 2** on this page.

2. A dermatologist will conduct an experiment to investigate the effectiveness of a new drug to treat acne. The dermatologist has recruited 36 pairs of identical twins. Each person in the experiment has acne and each person in the experiment will receive either the new drug or a placebo. After each person in the experiment uses either the new drug or the placebo for 2 weeks, the dermatologist will evaluate the improvement in acne severity for each person on a scale from 0 (no improvement) to 100 (complete cure).

   (a) Identify the treatments, experimental units, and response variable of the experiment.

- **Treatments:** the new drug and a placebo [①]
- **Experimental units:** the 72 people (each individual twin) in the experiment [②]
- **Response variable:** the improvement in acne severity (scale from 0 to 100) [③]

Each twin in the experiment has a severity of acne similar to that of the other twin. However, the severity of acne differs from one twin pair to another.

(b) For the dermatologist's experiment, describe a statistical advantage of using a matched-pairs design where twins are paired rather than using a completely randomized design.

Because the twins in each pair have similar acne severity, comparing the two twins in a pair removes the variation caused by different initial acne severity. This lets the dermatologist separate the effect of the drug from differences between people, so it is **easier to tell whether the new drug works** and the **estimate of the drug's effect is more precise [①] than with a completely randomized design [②]**, where twins from different pairs (with very different initial **acne** severity) could end up in different treatment groups [③].

1. 优势要落到**收集数据后的推断**上：更容易判断药是否有效、药效估计更精确、药和安慰剂的差别更容易区分。
2. 说明配对设计**更好**：用比较级（easier、clearer、more precise），或和完全随机设计做明确比较。
3. 有**语境**：drug、improvement、acne、twins 任一即可。

GO ON TO THE NEXT PAGE.

© 2022 College Board.
Visit College Board on the web: collegeboard.org.

Continue your response to **QUESTION 2** on this page.

(c) For the dermatologist's experiment, describe how the treatments can be randomly assigned to people using a matched-pairs design in which twins are paired.

Model answer

> For each pair of twins, label one twin "A" and the other "B." Flip a fair coin **[①]**. If it lands heads, twin A gets the new drug and twin B gets the placebo; if it lands tails, twin A gets the placebo and twin B gets the new drug **[②]**. **Repeat this for all 36 pairs of twins [③]**.

采分点（必须是在双胞胎对内部随机分配；三点全对 = E，对两点 = P）

1. 用一个随机过程（抛硬币、随机数生成器、洗匀的卡片），使每个人拿到新药和安慰剂的概率各 50%。
2. 写清如何用这个随机过程让**一个指定的双胞胎拿药、另一个拿安慰剂**：**每一种结果**（正面、反面）对应谁拿什么都要写。
3. 说明**每一对**双胞胎都重复这个过程。

GO ON TO THE NEXT PAGE.

© 2022 College Board.
Visit College Board on the web: collegeboard.org.